

a 5V phone charger adaptor can be used to power the circuit.

The mains circuit, shown in Fig.

3, is built around NodeMCU V3 and LCD LM016L (LCD1). Pin details of the NodeMCU are shown in Fig. 4.

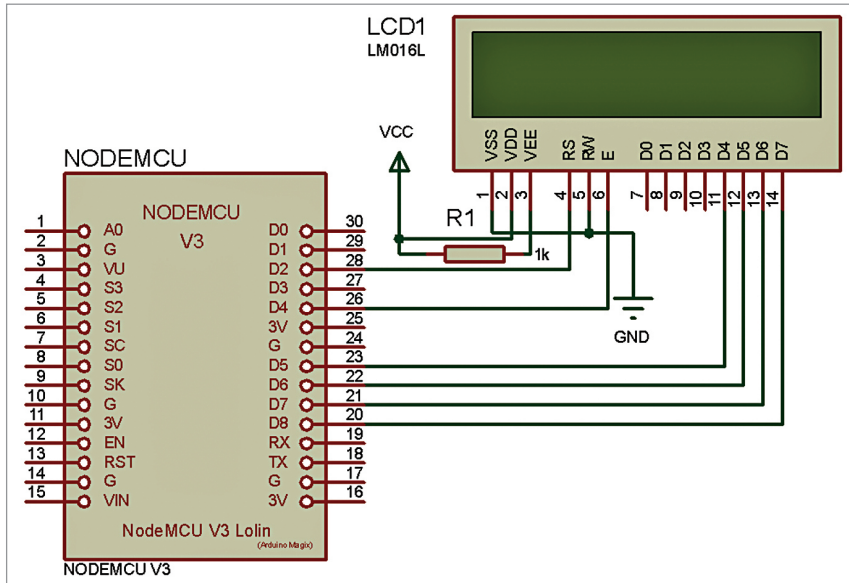


Fig. 3: The main circuit diagram

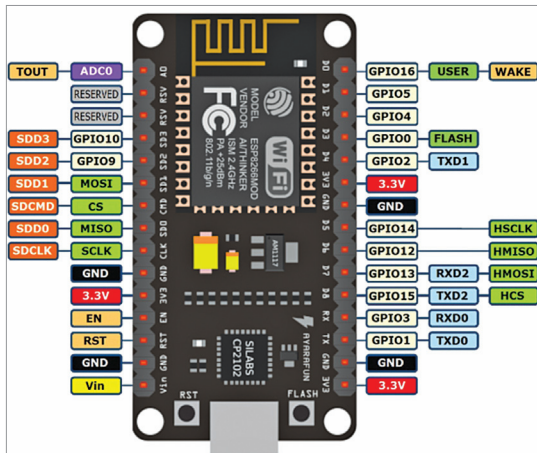


Fig. 4: Pin details of NodeMCU

The working principle of the project is as shown Fig. 1. The ESP8266 NodeMCU sends request packet to NTP server and then the NTP server sends a time stamp packet to ESP8266. The NodeMCU decodes the time stamp with the help of NTPClient.h header file along with User Datagram Protocol (UDP). After decoding the time stamp and day of the week in se-

rial port it is displayed on the 16x2 character LCD.

Installing NTP Client library

The easiest way to get time from an NTP server is by using an NTP Client from Arduino libraries. To install this library in your Arduino IDE, Navigate to the Sketch→Include Library→Manage Libraries... Wait for Library Manager to download libraries index and update the list of installed libraries.

Filter your search by typing *ntpclient*. There should be a couple of entries. Look for *NTPClient* by *Fabrice Weinberg*. Click on that entry, and then select Install, as shown in Fig. 5.

Software

The circuit requires the software program loaded into the internal memory of NodeMCU to operate. Arduino IDE 1.6.8 is used to compile and upload the program. The program makes the device communicate with NTP Application Platform when connected to the internet through an access point or Wi-Fi router.

To add board for ESP8266 NodeMCU in Arduino IDE, go to File→Preferences and paste the link http://arduino.esp8266.com/stable/package_esp8266com_index.json

Open Board Manager from Tools→Board Menu and install ESP8266 platform. (Do not forget to select your ESP8266 board from Tools→Board menu after installation.)

The sketch is at the heart of the system and carries out all major functions. For external header files needed for programming, use the above link.

After uploading the sketch, press RST button on your NodeMCU. You would get the current day and time every second, as shown in Fig. 6.

Let's take a quick look at the code to see how it works. First, we

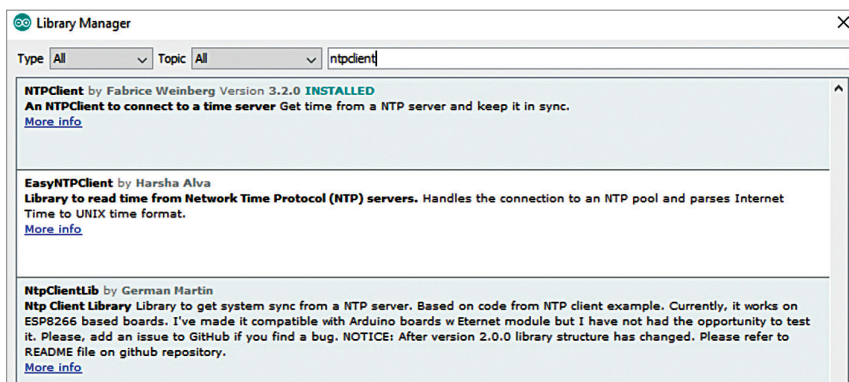


Fig. 5: NTPClient.h library installation steps